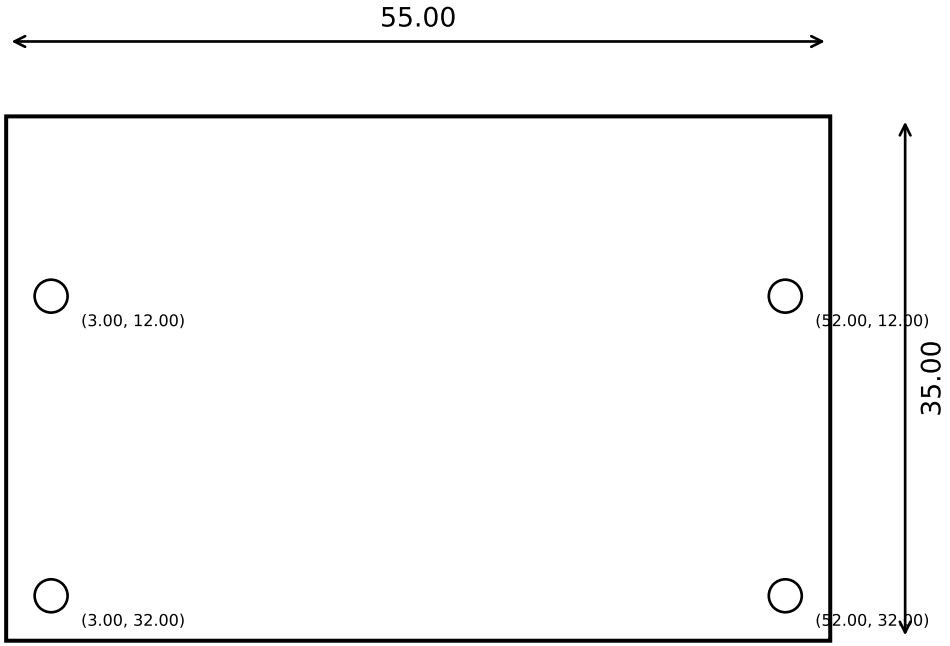
display-adapter — fabrication drawing



Finished thickness: 1.0 mm

2 layers, FR-4

1 oz copper, all layers

ENIG

Green mask / white silk

IPC Class 2

Electrical test: 100%

Layer order: F.Cu / B.Cu. Routed Edge.Cuts and drill files are authoritative. Nominal stackup is embedded in the KiCad board; equivalent FR-4 laminate construction is acceptable at the specified finished thickness.

Minimum track and copper clearance 0.15 mm; through via drill 0.25 mm, diameter 0.55 mm, annular ring 0.15 mm. Tented vias both sides, no via-in-pad. Preserve fine-pitch solder-mask webs (minimum 0.13 mm).

PCBWay to add assembly rails/panelization without changing individual outlines or mounting coordinates. Do not mirror bottom fabrication files. PTH and NPTH files are separate.

Origin in Gerbers, drills and placement: X=0, Y=lower board edge; positive Y upward. Drawing coordinates above use KiCad top-left Y-down for mechanical reference.

Stencil: stepped 0.15 mm at FH28 and 0.10–0.12 mm at 0402/TSSOP parts. Hirose paste apertures are already reduced in the supplied paste layers. No substitutions for connector suffixes or header tail lengths.

display-adapter — assembly instructions

Use assembly-F.pdf and assembly-B.pdf for reference designators, pin numbers and component orientation. Bottom drawing is mirrored to show the actual bottom view. Native KiCad placement rotations are retained in placement.csv.

All components in bom-pcbway-smt.csv are FIT. No DNP SMT components. Parts marked HAND in bom-all.csv are supplied/installed separately and must not appear in the SMT placement file.

Inspect every 0.5 mm connector contact for bridges, insufficient solder and coplanarity. Check soldermask webs and that no solder or un-tented via protrudes into the Hirose underside/mating keepout.

J1 DF23C-50DP-0.5V(92) is BOTTOM mounted. Viewed from the PCB top: pin 1 is lower left of the contact array, pin 2 above it, pin 49 lower right. Cross-check Smartwin rear-view Detail A before mating.

J2 FH28H is TOP mounted, mouth facing the upper board edge, pin 1 at the right end viewed from the top. Fit the cable with the specified same-side contact orientation.

U1 TRACO TSR 2-2433 is HAND soldered on TOP. Pin 1 = 24V in, pin 2 = GND, pin 3 = 3.3V out. Follow the pin-1 marking; the 12V converter is NOT interchangeable.

After soldering U1, clip its leads to at most 0.8 mm below the PCB, preserving the solder joints. Untrimmed module leads protrude below the nominal display mating plane. The STEP assembly model shows trimmed leads.

Support the PCB on four M2 insulating mounts and clamp the FFC/the ribbon to the enclosure. The display flex and DF23 must carry no cable or PCB weight. Maintain nominal 1.5 mm mating-board spacing at DF23.

First-board power, timing, touch and supply-sequencing tests are listed in README.md. They have not been performed.